Component Diagram – Layered Architecture: Academic Integrity, Regrade Appeals & Cohort Analytics

Automated Rubric Assignment Evaluator · Test Case 4 · PES1UG24AM165 (Mohammed Sinan M T)

«layer» Presentation Layer

«component»
Web Portal UI
Student: own report, case, appeal
Faculty Evaluator: queue, decisions, analytics

«layer» Business Layer

ISecureAPI
HTTPS / REST, TLS 1.2+

«component»
Access Control & API Gateway
Authentication · role check (Student / Faculty Evaluator) · per-student ownership filter
TLS 1.2+ · elevated-role guard on identified exports (NFR-001)

IScanJob
REST + job queue

«component»
Similarity Scan Service
Token-level structural scan
as async background job
FR-001 · NFR-002

IScanReport
internal API

ICaseMgmt
REST / JSON

«component»
Integrity Case Manager
Open case, evidence pack,
response window, decision
FR-002

INotify
event / SMTP

«component»
Notification Service
Alerts to students and
faculty evaluators
FR-002 · FR-003 · FR-004

INotify
event / SMTP

IAppealMgmt
REST / JSON

«component»
Regrade & Score Revision Manager
72 h window, review queue,
score revision + history
FR-003 · FR-004

IAnalyticsRefresh
internal API

IAnalytics
REST / JSON

«component»
Analytics & Export Service
Distribution, pass rate, heat map,
CSV export
FR-005 · NFR-002

«layer» Data Layer

IUserRoleStore
SQL / JDBC

IScanStore
SQL / JDBC

ICaseStore
SQL / JDBC

IAppealStore
SQL / JDBC

IReportQuery
SQL / JDBC

«component»
Database Component (Repository)
Submissions + archive · similarity reports · integrity cases · appeals · score history · audit log

Notation: ○ **ball = provided interface** ⊂ **socket = required interface** **(socket sits on the consumer, ball on the provider; label = interface + protocol)**

Example flow (regrade appeal): Web Portal UI → ISecureAPI → Access Control → IAppealMgmt → Regrade & Score Revision Manager → IAppealStore → Database;
then INotify → Notification Service (student told of decision) and IAnalyticsRefresh → Analytics & Export Service (dashboard and CSV recomputed).